# Semester Project: Smart Parking Platform

## Research

- Competitive Analysis

| Platform | Strengths | Weaknesses | Opportunities |
|---|---|---|---|
| ParkMobile | Easy meter payment; session extension in the app; garage reservations | Transaction fee added at checkout | Show all-in price before booking |
| PayByPhone | - Send reminders on the remaining parking hours<br>- Send receipts<br>- Wide coverage | - No spot reservation, no price transparency | Reservations with clear pricing |
| SpotHero | Large network of private garages, lots and valet; reservation | Fees added at checkout; drivers arrive to find the garage full or the pass refused, and must resolve it through customer support, with refunds taking days or issued as app credit | Return unused time as credit |
| ParkWhiz | Garage and event parking reservations; wide U.S. coverage | Passes rejected at facilities; slow refunds; app credit | Guarantee entry or an immediate alternative at no extra cost |

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| | | offered instead of refunds | |

- Differentiation Summary:

  The Smart Parking Platform differentiates itself through transparency in both pricing and reservations. First, the full price, including all fees and taxes, is displayed as a single total before booking, with an itemized breakdown available on request. Second, every reservation is guaranteed. If a driver arrives and cannot enter the garage because it is full, the pass is refused, or the entry system fails, the driver can report the issue in the app. The platform then immediately directs the driver to the nearest available garage at no additional cost and refunds the original payment to the original payment method.

## Vision and Scope

- Project Acquisition
  - Research by WE ARE $oftware ¢orp found two recurring problems with existing garage parking platforms. Fees are often added at checkout rather than shown upfront, and drivers with paid reservations are frequently unable to enter the garage because it is full, the pass is refused, or the entry system fails. When this happens, drivers must resolve the issue through customer support, often paying for parking elsewhere and waiting days for a refund. The company presented these findings and a proposed solution to the city transportation department, which then issued a formal Request for Proposal (RFP). WE ARE $oftware ¢orp was selected as the contractor.
- Company Introduction
  - WE ARE $oftware ¢orp is a software development studio specializing in consumer-facing mobility and civic technology solutions. The company is recognized for delivering secure, reliable, and user-centered applications.
- Project Overview
  - The Smart Parking Platform is a web and mobile application that connects drivers with partner parking garages. Drivers can locate, reserve, and pay for parking

spaces, while garage operators can manage facilities, rates, and occupancy. The platform does not own or operate parking facilities.

- The platform is built on transparency in pricing and reservations. Each price is displayed as a single all-inclusive total before booking, with an itemized breakdown available on request. If a driver with a valid reservation cannot enter the garage, the driver can report the issue in the app. The platform verifies that the driver is at the garage, directs the driver to the nearest available partner garage at no additional cost, and refunds the original payment to the original payment method. These costs are recovered from the garage that failed to honor the reservation, under the terms of its partner agreement.

# Software Requirements Specification (SRS)

Project: Smart Parking Platform

Version: 1.0

Date: 09.10.2026

## 1. Introduction

### 1.1 Purpose

This document defines the requirements for the Smart Parking Platform, a system for finding, reserving, and paying for garage parking with all-inclusive pricing and guaranteed entry.

### 1.2 Scope

The system covers user registration, parking search and reservation, all-inclusive price display, payment processing, guaranteed entry (reporting of entry failures, rebooking at a nearby garage, and refunds), notifications, receipts and history, the operator dashboard, and occupancy reporting. It does not cover street or metered parking, ownership or operation of parking facilities, installation of gate or camera hardware, payment processor backend infrastructure, or parking enforcement (citations, towing).

## 2. Overall Description

- **Users:** Drivers, parking facility operators, city transportation department staff, system administrators, finance and billing personnel
- **System Environment:** Web application and mobile application (iOS/Android), supported by a cloud-hosted backend

- **Constraints:** Must integrate with third-party mapping, navigation, and payment services; must comply with security and privacy requirements and city ordinances; fixed project timeline
- **Assumptions:** Users have internet-connected devices; partner garages provide real-time availability data; partner agreements require garages to honor reservations and cover the cost of rebooking when they do not

**3. Functional Requirements**

- **FR1:** The system shall allow drivers and operators to register, log in, and reset their passwords securely.
- **FR2:** The system shall display real-time parking availability on an interactive map.
- **FR3:** The system shall integrate with external mapping and navigation services to provide directions to the reserved garage.
- **FR4:** The system shall allow drivers to reserve a parking space at a partner garage for a chosen time window.
- **FR5:** The system shall allow drivers to cancel a reservation before its start time and receive a full refund to the original payment method.
- **FR6:** The system shall display a single all-inclusive price before booking, with an itemized breakdown available on request, and shall not charge any amount beyond that price for the reserved time.
- **FR7:** The system shall process digital payment at the time of booking and notify the driver if a payment fails.
- **FR8:** The system shall allow drivers to extend an active reservation if space is available.
- **FR9:** The system shall allow a driver with an active reservation to report an entry failure, such as a full garage, a refused pass, or an entry system failure.
- **FR10:** The system shall verify that the driver is located at the reserved garage when an entry failure is reported.
- **FR11:** Upon verification, the system shall rebook the driver at the nearest partner garage with available space at no additional cost and provide navigation to it.
- **FR12:** The system shall refund the original reservation payment to the original payment method, not as app credit, when an entry failure is verified.

- **FR13:** The system shall record each verified entry failure against the responsible garage for operator reporting and cost recovery.
- **FR14:** The system shall send notifications for reservation confirmation, upcoming start time, time remaining, rebooking confirmation, and refund issued.
- **FR15:** The system shall generate an itemized receipt for each reservation, including any rebooking or refund, and store reservation history for each user.
- **FR16:** The system shall provide operators with a dashboard to add, edit, and remove garages and to manage availability and rates.
- **FR17:** The system shall provide operators with occupancy, revenue, and entry-failure reports, including occupancy trends and graphs, with export capability.
- **FR18:** The system shall provide finance and billing personnel with reports on transactions, refunds, and rebooking costs recovered from garages, with export capability.
- **FR19:** The system shall provide city transportation department staff with read-only access to aggregated garage occupancy data.
- **FR20:** The system shall allow system administrators to manage user accounts and roles and to approve new partner garages and operators.

**4. Non-Functional Requirements**

- **NFR1:** The system shall support at least 10,000 concurrent users.
- **NFR2:** The system shall update availability data within 5 seconds of a status change.
- **NFR3:** The system shall maintain 99.9% uptime.
- **NFR4:** The system shall encrypt all payment and personal data in transit and at rest.
- **NFR5:** The system shall present a rebooking option within 60 seconds of a verified entry-failure report.
- **NFR6:** The system shall initiate the refund within 15 minutes of a verified entry failure.
- **NFR7:** The system shall collect the driver's location only when an entry failure is reported and shall retain it for no longer than 90 days.
- **NFR8:** The system shall process payments in compliance with the Payment Card Industry Data Security Standard (PCI DSS).
- **NFR9:** The web application shall support the two most recent versions of Chrome, Safari, Edge, and Firefox, and the mobile application shall support the two most recent major versions of iOS and Android.

- **NFR10:** The system shall comply with applicable city parking ordinances and with WCAG 2.1 AA accessibility standards.

**5. Use Cases**

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| Case 1 | Title: Register Driver Account<br>Actor: Driver<br>Precondition: App is installed<br>Steps:<br>1. Enter personal information and vehicle details<br>2. Verify email or phone number<br>3. Create a password<br>Postcondition: Driver account is created |
| Case 2 | Title: Register Operator Account<br>Actor: Parking Operator<br>Precondition: Operator has business and garage information available<br>Steps:<br>1. Enter company and garage details<br>2. Submit registration for review<br>3. System administrator approves the account<br>Postcondition: Operator account is active |
| Case 3 | Title: Log In<br>Actor: Driver, Parking Operator<br>Precondition: Account exists<br>Steps:<br>1. Enter credentials<br>2. System authenticates user<br>Postcondition: User is logged in |

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| Case 4 | Title: Reset Password<br>Actor: Driver, Parking Operator<br>Precondition: Account exists; user cannot log in<br>Steps:<br>1. Select "Forgot password"<br>2. Open the reset link sent by email<br>3. Enter and confirm a new password<br>Postcondition: Password is updated |
| Case 5 | Title: Search for Parking<br>Actor: Driver<br>Precondition: User is logged in<br>Steps:<br>1. Enter destination<br>2. View available garages on the map<br>Postcondition: Available garages and spaces are displayed |
| Case 6 | Title: View All-Inclusive Price<br>Actor: Driver<br>Precondition: Garage and time window are selected<br>Steps:<br>1. System calculates the total, including all fees and taxes<br>2. System displays the single total price<br>3. Driver taps the information icon to view the itemized breakdown (optional)<br>Postcondition: Driver sees the exact price to be charged |

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| Case 7 | Title: Reserve a Space<br>Actor: Driver<br>Precondition: Price is displayed<br>Steps:<br>1. Confirm the garage and time window<br>2. Select "Reserve"<br>Postcondition: Reservation is created and awaiting payment |
| Case 8 | Title: Make Payment<br>Actor: Driver<br>Precondition: Reservation is created<br>Steps:<br>1. Select payment method<br>2. Submit payment<br>3. System confirms payment<br>Postcondition: Payment is processed at the displayed price and the reservation is confirmed |
| Case 9 | Title: Handle Failed Payment<br>Actor: Driver<br>Precondition: Payment has been declined<br>Steps:<br>1. System notifies the driver and holds the space for 5 minutes<br>2. Driver selects a different payment method<br>3. Driver resubmits payment<br>Postcondition: Reservation is confirmed, or the space is released if payment is not completed |

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| Case 10 | Title: Cancel Reservation<br>Actor: Driver<br>Precondition: Reservation is confirmed and has not started<br>Steps:<br>1. Open the reservation<br>2. Select "Cancel"<br>3. Confirm cancellation<br>Postcondition: Reservation is cancelled and a full refund is issued to the original payment method |
| Case 11 | Title: Navigate to Garage<br>Actor: Driver<br>Precondition: Reservation is confirmed<br>Steps:<br>1. Select "Navigate"<br>2. App opens turn-by-turn directions to the garage entrance<br>Postcondition: Driver is routed to the reserved garage |
| Case 12 | Title: Receive Notification<br>Actor: Driver<br>Precondition: Driver has a reservation, rebooking, or refund in progress<br>Steps:<br>1. System detects an event (confirmation, upcoming start time, time remaining, rebooking confirmed, refund issued)<br>2. System sends an in-app notification<br>Postcondition: Driver is informed of the event |

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| Case 13 | Title: Extend Reservation<br>Actor: Driver<br>Precondition: Reservation is active and space is available after the current end time<br>Steps:<br>1. Select "Extend"<br>2. Review the updated all-inclusive price<br>3. Confirm and pay<br>Postcondition: Reservation end time is extended |
| Case 14 | Title: Report Entry Failure<br>Actor: Driver<br>Precondition: Reservation is active; driver cannot enter the garage<br>Steps:<br>1. Select "I can't get in"<br>2. Choose the reason (garage full, pass refused, entry system failure)<br>3. System verifies the driver's location at the reserved garage<br>Postcondition: Entry failure is verified |
| Case 15 | Title: Entry Failure Not Verified<br>Actor: Driver<br>Precondition: Driver reported an entry failure but is not located at the reserved garage<br>Steps:<br>1. System detects that the driver's location does not match the garage<br>2. System notifies the driver that the report cannot be verified<br>3. System provides a link to contact customer support<br>Postcondition: No rebooking or refund is issued; the reservation remains active |

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| Case 16 | Title: Rebook at Nearby Garage<br>Actor: System<br>Precondition: Entry failure is verified<br>Steps:<br>1. System finds the nearest partner garage with available space<br>2. System creates a new reservation at no additional cost to the driver<br>3. System sends the new reservation and starts navigation<br>Postcondition: Driver is rebooked at a nearby garage; if no garage is available, the driver is notified and receives a full refund |
| Case 17 | Title: Receive Refund<br>Actor: Driver<br>Precondition: Entry failure is verified<br>Steps:<br>1. System initiates a refund of the original payment to the original payment method<br>2. System sends a refund confirmation<br>Postcondition: Refund is issued; no app credit is used |
| Case 18 | Title: Record Entry Failure Against Garage<br>Actor: System<br>Precondition: Entry failure is verified<br>Steps:<br>1. System logs the failure with time, reason, and garage<br>2. System adds the rebooking cost to the garage's account for recovery<br>Postcondition: Failure appears in the operator and finance reports |

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| Case 19 | Title: View Itemized Receipt<br>Actor: Driver<br>Precondition: Reservation has ended, or a rebooking or refund has occurred<br>Steps:<br>1. Open the receipt<br>2. System displays the price breakdown and any rebooking or refund<br>Postcondition: Driver sees a full record of the transaction |
| Case 20 | Title: View Reservation History<br>Actor: Driver<br>Precondition: User is logged in<br>Steps:<br>1. Open the history tab<br>2. View past reservations<br>Postcondition: Past reservations are displayed |
| Case 21 | Title: Manage Garage and Rates<br>Actor: Parking Operator<br>Precondition: Operator is logged into the dashboard<br>Steps:<br>1. Open the garage panel<br>2. Add, edit, or remove a garage, or update availability or rates<br>3. Save changes<br>Postcondition: Garage information is updated system-wide |

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| Case 22 | Title: View Occupancy and Entry-Failure Reports<br>Actor: Parking Operator<br>Precondition: Operator is logged into the dashboard<br>Steps:<br>1. Open the reports panel<br>2. Select a date range<br>3. View occupancy trends, graphs, revenue, and entry failures, or export the report<br>Postcondition: Report is displayed or exported |
| Case 23 | Title: Export Financial Report<br>Actor: Finance and Billing Personnel<br>Precondition: User is logged in with finance access<br>Steps:<br>1. Open the finance reports panel<br>2. Select a date range<br>3. Export transactions, refunds, and rebooking costs recovered<br>Postcondition: Financial report is exported |
| Case 24 | Title: View City Occupancy Data<br>Actor: City Transportation Department Staff<br>Precondition: User is logged in with read-only city access<br>Steps:<br>1. Open the city data panel<br>2. Select an area and date range<br>3. View aggregated occupancy across partner garages<br>Postcondition: Aggregated occupancy data is displayed |

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| Case 25 | Title: Manage User Accounts<br>Actor: System Administrator<br>Precondition: Administrator is logged in<br>Steps:<br>1. Search for a user or pending operator<br>2. Approve, update the role of, or deactivate the account<br>3. Save changes<br>Postcondition: Account status and role are updated |

**Driver**

**User Story 1: Create an Account**

As a driver, I need to create an account with my personal and vehicle information.

Acceptance Criteria:

- Driver can enter name, email, phone number, and vehicle details.
- System sends a verification code to the email or phone number.
- Account is created only after verification.
- Password must meet security requirements.

**User Story 2: Log In and Reset Password**

As a driver, I need to log in securely and reset my password if I forget it.

Acceptance Criteria:

- Driver can log in with email and password.
- Driver can request a password reset link by email.
- Reset link expires after 30 minutes.
- Driver can log in with the new password.

**User Story 3: Find Parking**

As a driver, I need to find garages with available space near my destination.

Acceptance Criteria:

- Driver can enter a destination.
- Map displays nearby partner garages.
- Each garage shows current availability.
- Availability updates within 5 seconds of a change.

**User Story 4: See the Full Price**

As a driver, I need to see the full price before I book.

Acceptance Criteria:

- System displays one total price, including all fees and taxes.
- Driver can tap an information icon to view the itemized breakdown.
- Amount charged equals the displayed total.

**User Story 5: Reserve and Pay**

As a driver, I need to reserve a space and pay for it in the app.

Acceptance Criteria:

- Driver can select a garage and time window.
- Driver can pay with a saved or new payment method.
- System confirms the reservation after payment succeeds.
- If payment fails, the driver is notified and the space is held for 5 minutes.

**User Story 6: Cancel a Reservation**

As a driver, I need to cancel a reservation before it starts.

Acceptance Criteria:

- Cancel option is available until the reservation start time.
- Driver receives a full refund to the original payment method.
- Space becomes available to other drivers.

**User Story 7: Navigate to the Garage**

As a driver, I need directions to the garage entrance.

Acceptance Criteria:

- Driver can start navigation from the reservation.
- Directions lead to the garage entrance, not just the street address.
- Navigation opens in the driver's mapping app.

**User Story 8: Receive Notifications**

As a driver, I need to be notified about my reservation.

Acceptance Criteria:

- Driver receives notifications for confirmation, upcoming start time, and time remaining.
- Driver receives notifications when a rebooking or refund occurs.
- Driver can turn notification types on or off.

**User Story 9: Extend a Reservation**

As a driver, I need to extend my reservation if I need more time.

Acceptance Criteria:

- Extend option appears only if space is available.
- Updated all-inclusive price is shown before confirming.
- New end time is confirmed after payment.

**User Story 10: Report That I Can't Get In**

As a driver, I need to report that I cannot enter the garage and be rebooked right away.

Acceptance Criteria:

- "I can't get in" option is available during an active reservation.
- Driver can select a reason: garage full, pass refused, or entry system failure.
- System verifies the driver is at the reserved garage.
- Rebooking option at a nearby garage appears within 60 seconds, at no additional cost.
- If no garage is available, the driver receives a full refund.

**User Story 11: Get My Money Back**

As a driver, I need my refund returned to my card, not as app credit.

Acceptance Criteria:

- Refund goes to the original payment method.
- Refund is initiated within 15 minutes of a verified entry failure.
- Driver receives a refund confirmation.

**User Story 12: View Receipts and History**

As a driver, I need to view my receipts and past reservations.

Acceptance Criteria:

- Each receipt shows the price breakdown.
- Receipts include any rebooking or refund.
- History lists all past reservations by date.

**Parking Operator**

**User Story 13: Register My Garage Business**

As a parking operator, I need to register my company and garages on the platform.

Acceptance Criteria:

- Operator can submit company and garage details.

- Registration is reviewed by a system administrator.
- Operator receives an email when the account is approved.

**User Story 14: Manage Garages and Rates**

As a parking operator, I need to manage my garages, availability, and rates.

Acceptance Criteria:

- Operator can add, edit, and remove garages.
- Operator can update available spaces and rates.
- Changes appear system-wide immediately after saving.

**User Story 15: View Reports**

As a parking operator, I need to view occupancy, revenue, and entry-failure reports.

Acceptance Criteria:

- Operator can select a date range.
- Reports include occupancy trends and graphs.
- Reports list each entry failure with time and reason.
- Operator can export reports.

**Finance and Billing Personnel**

**User Story 16: Export Financial Reports**

As a finance and billing staff member, I need to export reports on transactions, refunds, and rebooking costs.

Acceptance Criteria:

- Staff can select a date range.
- Report includes transactions, refunds, and rebooking costs recovered from garages.
- Report can be exported.

**City Transportation Department Staff**

**User Story 17: View City Occupancy Data**

As a city transportation department staff member, I need to view occupancy data across partner garages.

Acceptance Criteria:

- Staff can select an area and date range.
- Data is aggregated and contains no personal driver information.
- Access is read-only.

**System Administrator**

**User Story 18: Manage User Accounts**

As a system administrator, I need to manage user accounts and approve new operators.

Acceptance Criteria:

- Administrator can search for any user.
- Administrator can approve, update the role of, or deactivate an account.
- Changes take effect immediately.

# WBS

**1. Authentication**

- 1.1 Registration
    - 1.1.1 Driver registration (personal and vehicle information)
    - 1.1.2 Operator registration
    - 1.1.3 Email and phone verification
- 1.2 Login
    - 1.2.1 Credential validation
    - 1.2.2 Session token handling
- 1.3 Session Management
    - 1.3.1 Timeout and logout
    - 1.3.2 Password reset

**2. User/Operator Setup**

- 2.1 Driver Profile
    - 2.1.1 Personal information management
    - 2.1.2 Vehicle management
    - 2.1.3 Saved payment methods
    - 2.1.4 Notification preferences
- 2.2 Operator Setup
    - 2.2.1 Company profile
    - 2.2.2 Operator approval workflow
- 2.3 Roles and Permissions
    - 2.3.1 Role management (driver, operator, finance, city, administrator)

    - 2.3.2 Account deactivation

**3. Garage Monitoring**

- 3.1 Real-Time Availability
    - 3.1.1 Garage data feed integration
    - 3.1.2 Occupancy status updates
- 3.2 Interactive Map
    - 3.2.1 Garage search by destination
    - 3.2.2 Map display with live availability
    - 3.2.3 Navigation to garage entrance (mapping service integration)
- 3.3 Garage Management Dashboard
    - 3.3.1 Add, edit, and remove garages
    - 3.3.2 Availability and rate management

**4. Payment**

- 4.1 Reservation Flow
    - 4.1.1 Garage and time window selection
    - 4.1.2 Reservation creation
    - 4.1.3 Reservation cancellation
    - 4.1.4 Reservation extension
- 4.2 All-Inclusive Pricing
    - 4.2.1 Total price calculation (fees and taxes)
    - 4.2.2 Itemized breakdown display
- 4.3 Payment Processing
    - 4.3.1 Payment gateway integration
    - 4.3.2 Failed payment handling
    - 4.3.3 Refund to original payment method
- 4.4 Guaranteed Entry
    - 4.4.1 Entry failure reporting
    - 4.4.2 Location verification
    - 4.4.3 Automatic rebooking at a nearby garage
    - 4.4.4 Cost recovery from garage
- 4.5 Receipts and History

        - 4.5.1 Itemized receipt generation
        - 4.5.2 Reservation history
- 4.6 Notifications and Alerts
    - 4.6.1 Reservation notifications
    - 4.6.2 Rebooking and refund notifications
    - 4.6.3 Email delivery

**5. Reporting**

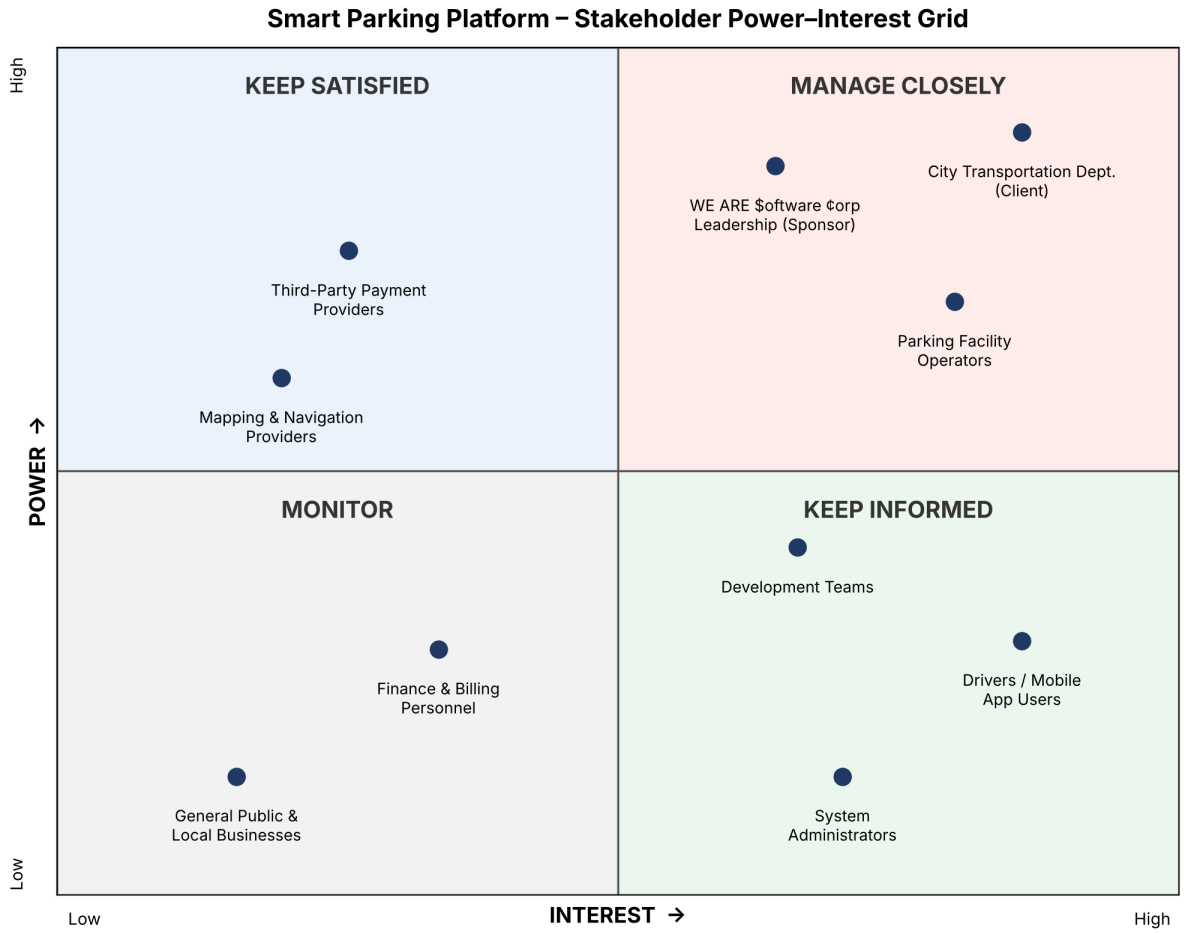
- 5.1 Operator Reports
    - 5.1.1 Occupancy trends and graphs
    - 5.1.2 Revenue reports
    - 5.1.3 Entry-failure reports
    - 5.1.4 Report export
- 5.2 Financial Reports
    - 5.2.1 Transactions and refunds report
    - 5.2.2 Rebooking cost recovery report
    - 5.2.3 Report export
- 5.3 City Data Access
    - 5.3.1 Aggregated occupancy view
    - 5.3.2 Read-only access control

**6. Testing and Deployment**

- 6.1 Quality Assurance
    - 6.1.1 Test plan and test cases
    - 6.1.2 Code review process
- 6.2 Quality Control
    - 6.2.1 Integration testing
    - 6.2.2 Security and performance testing
    - 6.2.3 User acceptance testing
- 6.3 Deployment
    - 6.3.1 Production environment setup
    - 6.3.2 App store release (iOS and Android)
    - 6.3.3 Operator onboarding and training

# Stakeholder Management

The Power-Interest Grid below classifies the project's stakeholders by their power and interest and maps out how each group will be managed.




| Stakeholder | Power | Interest | Quadrant | Management Strategy |
|---|---|---|---|---|
| City Transportation Department (Client) | High | High | Manage Closely | Monthly progress reviews; approves scope changes and milestones |
| WE ARE $oftware ¢orp | High | High | Manage Closely | Biweekly status reports; budget |

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| Leadership (Sponsor) | | | | and schedule decisions |
| Parking Facility Operators | High | High | Manage Closely | Partner agreements, onboarding sessions, and regular check-ins; required for availability data and guaranteed entry |
| Third-Party Payment Providers | High | Low | Keep Satisfied | Meet integration and compliance requirements; contact on contract or API changes |
| Mapping & Navigation Service Providers | High | Low | Keep Satisfied | Follow API terms and usage limits; monitor pricing changes |
| Drivers / Mobile App Users | Low | High | Keep Informed | Beta testing, release notes, and in-app announcements |
| Development Teams | Medium | High | Keep Informed | Daily stand-ups, sprint reviews, and retrospectives |
| System Administrators | Low | High | Keep Informed | Technical documentation, |

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|---|---|---|---|---|
| | | | | release schedules, and training before launch |
| Finance & Billing Personnel | Low | Medium | Monitor | Monthly reports on transactions, refunds, and rebooking costs |
| General Public & Local Businesses | Low | Low | Monitor | Public announcements at launch |

## Project Charter

| PROJECT CHARTER | | | |
|---|---|---|---|
| **Project Title** | Smart Parking Platform | **Project Manager** | An Tran |
| **Project Start Date** | 08/31/2026 | **Project End Date** | 08/31/2027 |
| **Project Sponsor** | WE ARE $oftware ¢orp Leadership | **Client** | City Transportation Department |
| **Business Need** | | | |
| Drivers lose time and money when garage fees are added at checkout and when paid reservations are not honored at the garage entrance. Parking operators need better tools to manage occupancy, pricing, and reservations. This project was initiated to deliver a web and mobile platform that connects drivers with partner parking garages through all-inclusive pricing and guaranteed entry. | | | |
| **Project Scope** | | **Deliverables** | |
| In scope: registration and authentication, real-time availability, interactive map, | | 1. Web application<br>2. iOS and Android mobile applications | |

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| reservations, digital payment, receipts and history, operator dashboard, occupancy reporting, notifications, mapping and navigation integration, all-inclusive pricing, and guaranteed entry.<br>Out of scope: street and metered parking, garage ownership or operation, gate or camera hardware installation, payment processor backend, and parking enforcement. | 3. Operator dashboard and occupancy reports<br>4. All-inclusive price display<br>5. Guaranteed entry: rebooking and refunds<br>6. Project management plan |
| **Risks and Issues** | **Assumptions/Dependencies** |
| 1. Garage operators may decline partner agreements.<br>2. Third-party API changes or outages may disrupt the service.<br>3. Abuse of the entry-failure report feature. | 1. Partner garages will provide real-time availability data and honor reservations.<br>2. The platform depends on third-party payment and mapping services.<br>3. The system must comply with security, privacy, and city ordinance requirements. |

**Financials**

Funded by WE ARE $oftware ¢orp under an unlimited theoretical development budget. A detailed cost plan will be developed during the planning phase.

**Milestones Schedule**

| Milestone | Target Completion Date | Actual Date |
|---|---|---|
| Project Kickoff | 08/31/2026 | 08/31/2026 |
| Project Plan Approved | 12/04/2026 | |
| Design Complete | 01/03/2027 | |
| Core Booking Complete | 04/25/2027 | |
| Feature Complete | 06/20/2027 | |

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|---|---|---|---|
| Beta Release | | 07/11/2027 | |
| Public Launch | | 08/09/2027 | |
| Project Closed | | 08/31/2027 | |
| **Project Team** | | **Approval/Review Committee** | |
| **Project Manager** | An Tran | **Sponsor** | WE ARE $oftware ¢orp Leadership |
| **Product Owner** | TBD | **Client** | City Transportation Department |
| **Scrum Master** | TBD | **Finance Manager** | Finance & Billing Lead |
| **Team Members** | Mobile, Web, Backend/API, Mapping & Location, Payment Integration, and QA & Testing Teams | | |

## Team Charter

| **Mission** | **Scope** | **Goals** |
|---|---|---|
| To plan, build, and launch the Smart Parking Platform on schedule, delivering all-inclusive pricing and | Within: design, development, testing, and deployment of the web and mobile applications, operator dashboard, and | 1. Launch by August 9, 2027 with all required features.<br>2. Deliver each sprint's committed backlog items on time. |

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| guaranteed entry to drivers and partner garages. | payment and mapping integrations.<br>Outside: garage operations, gate or camera hardware, payment processor backend, and parking enforcement. | 3. Catch defects early through continuous code review and testing. |
| **Strengths & Skills** | **Roles & Responsibilities** | **Metrics for Success** |
| A cross-functional team with skills in iOS and Android development, web development, backend and cloud services, mapping and location services, payment integration and security compliance, and quality assurance and test automation, led by a project manager experienced in Agile practices. | Project Manager: plans, tracks progress, manages stakeholders and risks.<br>Product Owner: defines and prioritizes the backlog.<br>Scrum Master: facilitates ceremonies and removes blockers.<br>Mobile, Web, Backend/API, Mapping & Location, and Payment Integration Teams: build their components.<br>QA & Testing Team: test plans, code reviews, and testing. | At least 90% of committed stories completed each sprint.<br>Milestones met on their target dates.<br>Zero critical defects at launch.<br>Charged price matches displayed price on 100% of bookings.<br>Verified entry failures resolved within 60 seconds. |
| **Values** | **Norms** | **FUN!** |
| Transparency, accountability, respect, and quality. Mistakes are discussed openly without blame (psychological safety). | Daily 15-minute stand-ups; Jira updated daily.<br>Major decisions are democratic: team input is gathered before the Project Manager decides; in urgent | Sprint review demos celebrate completed work.<br>Monthly team lunch.<br>Celebrations at major milestones such as Beta Release and Public Launch. |

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| | situations the Project Manager may decide authoritatively.<br>Conflicts are resolved first through collaboration, then compromise facilitated by the Scrum Master, then a final decision by the Project Manager. | |
| **Communications Plan and Coordination Tools** | | |
| Tools: Microsoft Teams for messaging, Jira for task tracking, GitHub for source code and documentation.<br>Meetings: daily stand-ups; sprint planning, review, and retrospective every two weeks; monthly stakeholder progress meetings.<br>Messages are answered within one business day; urgent issues are raised immediately in Teams. | | |